When Average Calibration Fails:

Site-Conditional Federated Conformal Risk Control

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Conformal risk control certifies the missed-tumour rate, on average

What CRC guarantees (FNR loss, B = 1)

$$C_\lambda(X) = \{\, v : f(X)_v \geq 1 - \lambda \,\}$$

$$\hat{\lambda} = \inf\left\{\lambda : \hat{R}(\lambda) + \frac{B}{n+1} \leq \alpha\right\}$$

$$\Rightarrow \ \mathbb{E}\left[\ell\left(C_{\hat{\lambda}}(X_{\text{test}}), Y_{\text{test}}\right)\right] \leq \alpha$$

Distribution-free · finite-sample · any segmentation model

In a federation

- Calibration volumes stay at K hospitals
- Standard practice: pool every site's calibration scores into one threshold
- The guarantee still holds, on average

On average over what?

Pooled calibration holds on average, yet 8 of 20 hospitals miss the target

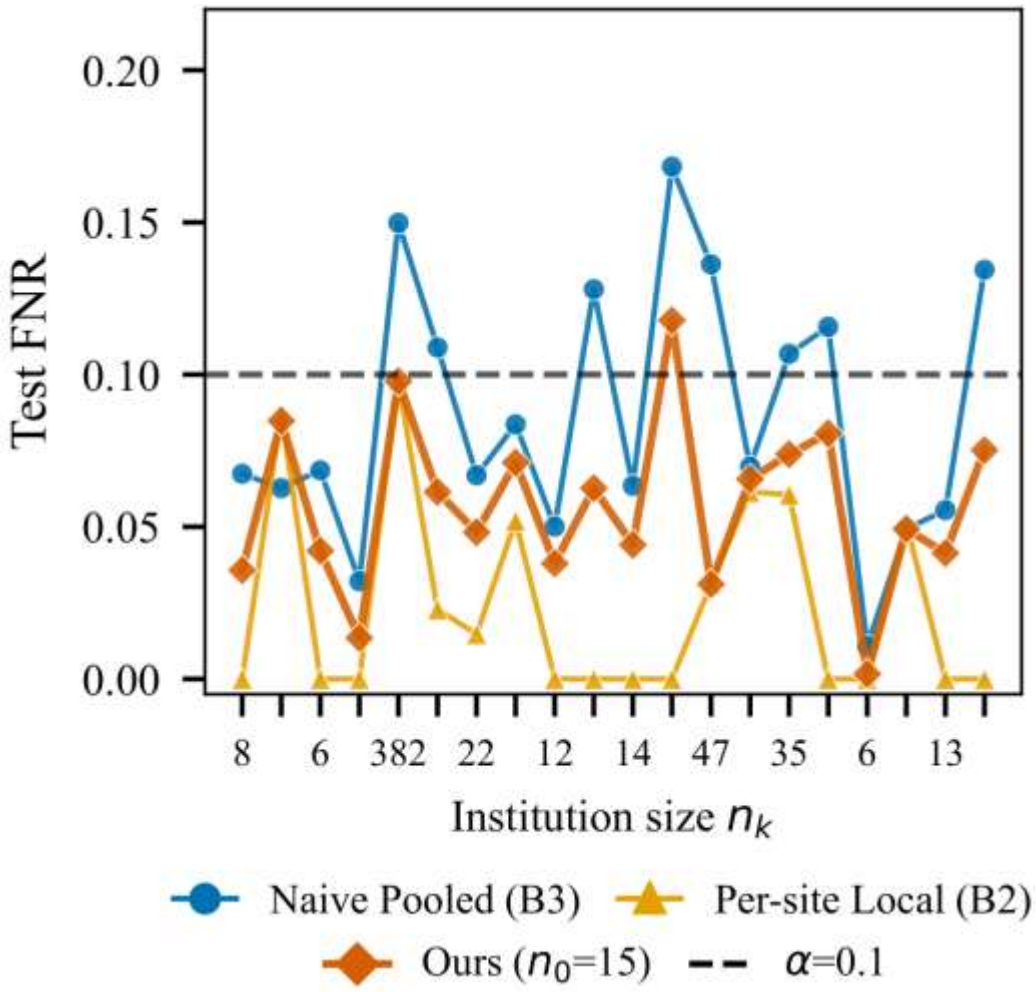


Fig. 3 (paper), left panel: per-institution test FNR, FeTS-2022, one calibration split

8 / 20 hospitals exceed α = 0.10 under pooled calibration: 40%

0.178 worst-site FNR, nearly double the target

382 patients at one failing hospital: 30% of all data

Mean over 3 calibration/test splits (Table 1)

The pooled number says calibrated. The hospitals say otherwise.

Pooling silently averages over patients, not over hospitals

Pooled scores: patient-uniform risk

$$R_{\text{micro}}(\lambda) = \sum_k \frac{n_k}{N} R_k(\lambda)$$

Large hospitals dominate

Every hospital equal: site-uniform risk

$$R_{\text{macro}}(\lambda) = \sum_k \frac{1}{K} R_k(\lambda)$$

Each institution counts the same

The two objectives differ by exactly one term (Eq. 1):

$$R_{\text{micro}}(\lambda) - R_{\text{macro}}(\lambda) = \frac{\text{Cov}_k\left(n_k,\ R_k(\lambda)\right)}{\bar{n}}$$

Whenever size correlates with risk, the aggregation weights decide whose risk is controlled.

Same four bytes per hospital, different weights: 8.3 versus 1.7 violations

Each hospital computes its own CRC threshold $\hat{\lambda}_k$ and sends only that number. The server averages:

Server aggregate	Sent per site	Violations / 20 ↓	Worst FNR ↓	Stretch ↓
Size-weighted: $\sum_k (n_k/N)\,\hat{\lambda}_k$	1 number	**8.3 ± 2.9**	0.186	1.4×
Unweighted: $(1/K)\sum_k \hat{\lambda}_k$	1 number	**1.7 ± 0.9**	0.113	2.3×
For reference, B3: naive pooled	all scores	8.0 ± 2.4	0.178	1.5×

Only the weights changed: same statistic, same communication.
Why: the finite-sample correction pushes small, calibration-poor hospitals to conservative thresholds; size weighting silences exactly those hospitals.

The failure is about whose risk the server optimises, not what hospitals share.

The two obvious fixes trade one failure for another

Calibrate each hospital alone (B2)

With n□ = 5 volumes, the correction alone exceeds α:

$$\frac{B}{n_k+1} = \frac{1}{5+1} \approx 0.17 \; > \; \alpha = 0.10$$

83×

average prediction-set size (1.3/20 violations)

“Calibration poverty”: small hospitals return the whole volume

Optimise set size (budget allocation)

Minimise total stretch subject to marginal risk ≤ α:

$$\min_{\lambda_1,\ldots,\lambda_K} \sum_k S_k(\lambda_k) \quad \text{s.t.} \quad \sum_k p_k L_k(\lambda_k) \le \alpha$$

12.3 / 20

violations; worst-site FNR 0.351

The optimiser meets the average by moving risk onto small, hard hospitals

The objective we optimise decides who bears the error.

Borrow strength from the federation while keeping site-specific thresholds

Each hospital

Sends its risk curve R̂ₖ(λ)

200 thresholds, ≈ 0.8 KB

No images, masks or per-volume scores

R̂ₖ(λ) →

Server

Shrinks each curve toward the federation's curve

One weight per hospital: wₖ

λ̂ₖ →

Each hospital

Receives its threshold λ̂ₖ

Site-specific prediction sets

$$\hat{R}_k^{\text{shrink}}(\lambda) = w_k\,\hat{R}_k(\lambda) + (1 - w_k)\,\hat{R}_{\text{global}}(\lambda) + \text{corr}_k$$

$$w_k = \frac{n_k}{n_k + n_0}$$

$n_0 \to 0$: per-site CRC

$n_0 \to \infty$: pooled CRC

n_0 is an empirical-Bayes prior strength; corrₖ is blended the same way

For a hard hospital, the shrunk curve crosses α between pooled and local calibration

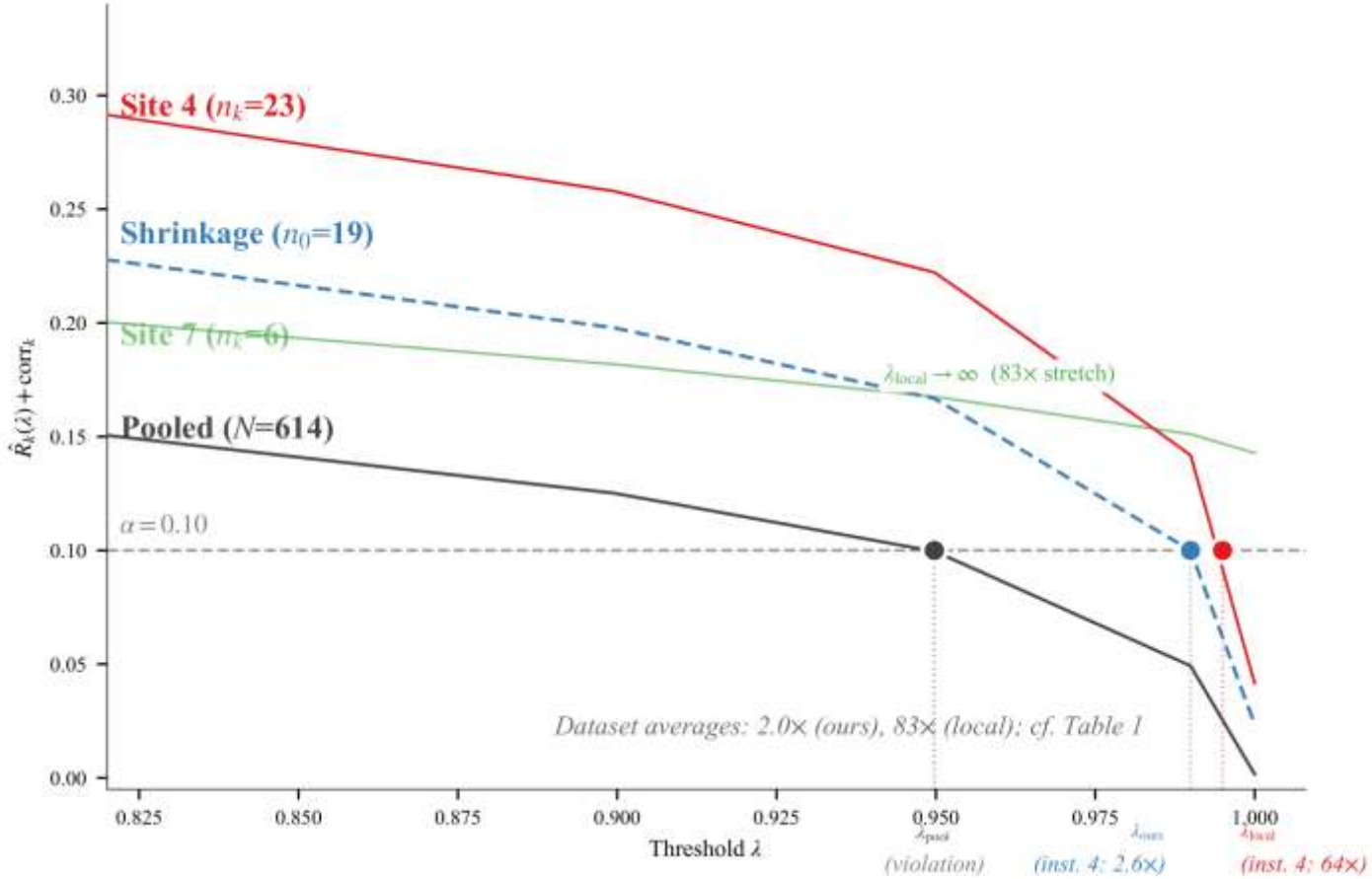


Fig. 2 (paper): one calibration split (seed 42)

- **Pooled** curve crosses α at $\hat{\lambda}_{pool} \approx 0.95$
- **Hard site 4** ($n_4 = 23$) is still well above α there: the violation
- **Its local threshold** costs a set 64× the tumour
- **Shrinkage** ($n_0 = 19$) crosses in between: 2.6× for site 4
- **Easy site 7** ($n_7 = 6$) never crosses in range: calibration poverty

Shrinkage reaches 2.7/20 violations at 2.0× mean stretch, versus 83.2× for per-site CRC

FeTS-2022, 20 institutions, α = 0.10, 3 seeds. Violations: mean ± std; worst FNR and stretch (set volume / tumour volume): mean.

Method	Violations ↓	Worst FNR ↓	Stretch ↓
B3: naive pooled	8.0 ± 2.4	0.178	1.5×
B2: per-site local	1.3 ± 1.2	0.111	83.2×
Unweighted λ average	1.7 ± 0.9	0.113	2.3×
Ours (n_0 = 19, LOSO)	2.7 ± 1.7	0.125	**2.0×**

Table 1 (paper), selected rows; full table in backup

- **97.6% lower** mean stretch than per-site CRC (2.0× vs 83.2×)
- **Without corr:** 8.0–9.3 violations
- **n_0 = 9:** 1.3 violations at 28.8×

Shrinkage does not beat the unweighted average; it adds per-site thresholds and a dial.

Before reporting a federated guarantee, ask whose risk it controls

1 Quantified: on real multi-institutional data, pooled CRC is calibrated on average but fails 8 of 20 hospitals

2 Identified: aggregation weights decide whose coverage is protected: same 4 bytes, 8.3 vs 1.7 violations

3 Proposed: risk-curve shrinkage, one dial n_0 and ≈ 0.8 KB per site: 2.7/20 violations at 2.0× mean stretch

Paper

Code

arXiv:2606.20115
github.com/NafisFuadShahid/Fed-CRC-Seg

Poster #4

Limitations: per-site thresholds are validated empirically, not guaranteed; one model, one dataset.

Backup: n_0 traces the coverage–efficiency frontier; LOSO selects $n_0 = 19$

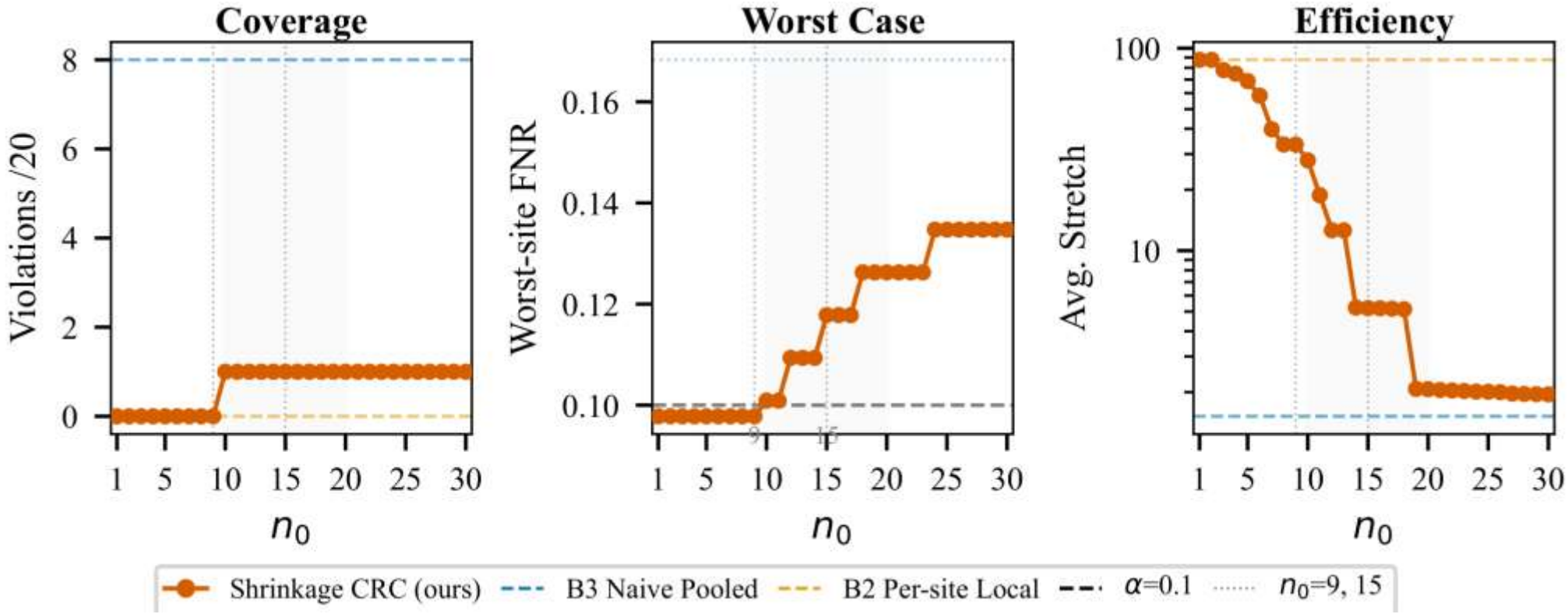


Fig. 4 (paper): effect of n_0 on violations (left), worst-site FNR (centre) and average stretch (right); shaded $n_0 \in [10, 20]$; one calibration split (seed 42), so values differ from the 3-seed means in Table 1

- Post-hoc leave-one-site-out analysis: hold out a site, rebuild the global curve from the other K−1, threshold the held-out site with its own calibration curve, test on its test split
- $n_0 = 19$: lowest average stretch with ≤ 3 mean violations

Backup: ablations and robustness to α

Ablation	n_o	Viol.	W-FNR	Stretch
No corr	9	9.3 ± 2.6	0.181	1.5×
No corr	15	8.3 ± 2.6	0.173	1.5×
$\hat{\lambda}$fed (Prop. 1)	9	0.0 ± 0.0	0.050	67.3×
$\hat{\lambda}$fed (Prop. 1)	15	0.3 ± 0.5	0.071	46.7×
Grid G = 50	15	2.3 ± 1.9	0.116	35.7×
Grid G = 500	15	3.3 ± 1.9	0.122	3.0×

Table 2 (paper): α = 0.10, mean over 3 seeds

- Across $\alpha \in \{0.05, 0.10, 0.15, 0.20\}$: naive pooling violates 5–8 sites at every α; ours (n_o = 9) 0–1
- Budget allocation with a per-site cap $\delta = 0.03$: 2.7 violations but 71.6× stretch, worse than shrinkage on both axes
- G = 200 is stable; G = 50 is too coarse

Backup: Algorithm 1 and the marginal safeguard (Proposition 1)

Algorithm 1 Shrinkage-Based Federated CRC

Require: Sites $k \in [K]$; target α; grid Λ; prior strength n_0

1: Each site k: compute $\hat{R}_k(\lambda)$ for $\lambda \in \Lambda$; enforce monotonicity via cumulative minimum over increasing λ; send to server
2: Server: $\hat{R}_{\text{global}}(\lambda) \leftarrow \frac{1}{N} \sum_k n_k \hat{R}_k(\lambda)$
3: **for** each site k **do**
4: Compute $\hat{R}_k^{\text{shrink}}$ (Eq. (2)); $\hat{\lambda}_k \leftarrow \inf\{\lambda : \hat{R}_k^{\text{shrink}}(\lambda) \leq \alpha\}$
5: **end for**
6: Broadcast $\hat{\lambda}_k$ to site k

Algorithm 1 (paper)

Proposition 1. If $\hat{\lambda}_{\text{pool}}$ satisfies the marginal guarantee, then for any per-site thresholds:

$$\hat{\lambda}_{\text{fed}} = \max\{\hat{\lambda}_{\text{pool}}, \max_k \hat{\lambda}_k\} \Rightarrow \mathbb{E}\left[\ell(C_{\hat{\lambda}_{\text{fed}}})\right] \leq \alpha$$

Proof: $\hat{\lambda}_{\text{fed}} \geq \hat{\lambda}_{\text{pool}}$ and the loss is non-increasing in λ. Deployed globally at $n_0 = 9$: 0 violations but 67× stretch, so per-site thresholds are used in practice.

Backup: full results (Table 1)

FeTS-2022, 20 institutions, $\alpha = 0.10$, 3 seeds. Violations: mean ± std; worst FNR and stretch: mean.

Method	Violations ↓	Worst FNR ↓	Stretch ↓
B3: Naive Pooled	8.0 ± 2.4	0.178	1.5×
B2: Per-site Local	1.3 ± 1.2	0.111	83.2×
Weighted λ Agg.	8.3 ± 2.9	0.186	1.4×
Unweighted λ Agg.	1.7 ± 0.9	0.113	2.3×
Budget Alloc. (uncapped)	12.3 ± 0.9	0.351	1.4×
Budget Alloc. ($\delta = 0.03$)	2.7 ± 1.2	0.142	71.6×
Ours ($n_o = 9$)	1.3 ± 1.2	0.112	28.8×
Ours ($n_o = 15$)	2.7 ± 1.7	0.119	4.4×
Ours ($n_o = 19$, LOSO)	2.7 ± 1.7	0.125	**2.0×**

Table 1 (paper)

Backup: protocol overview (paper Fig. 1)

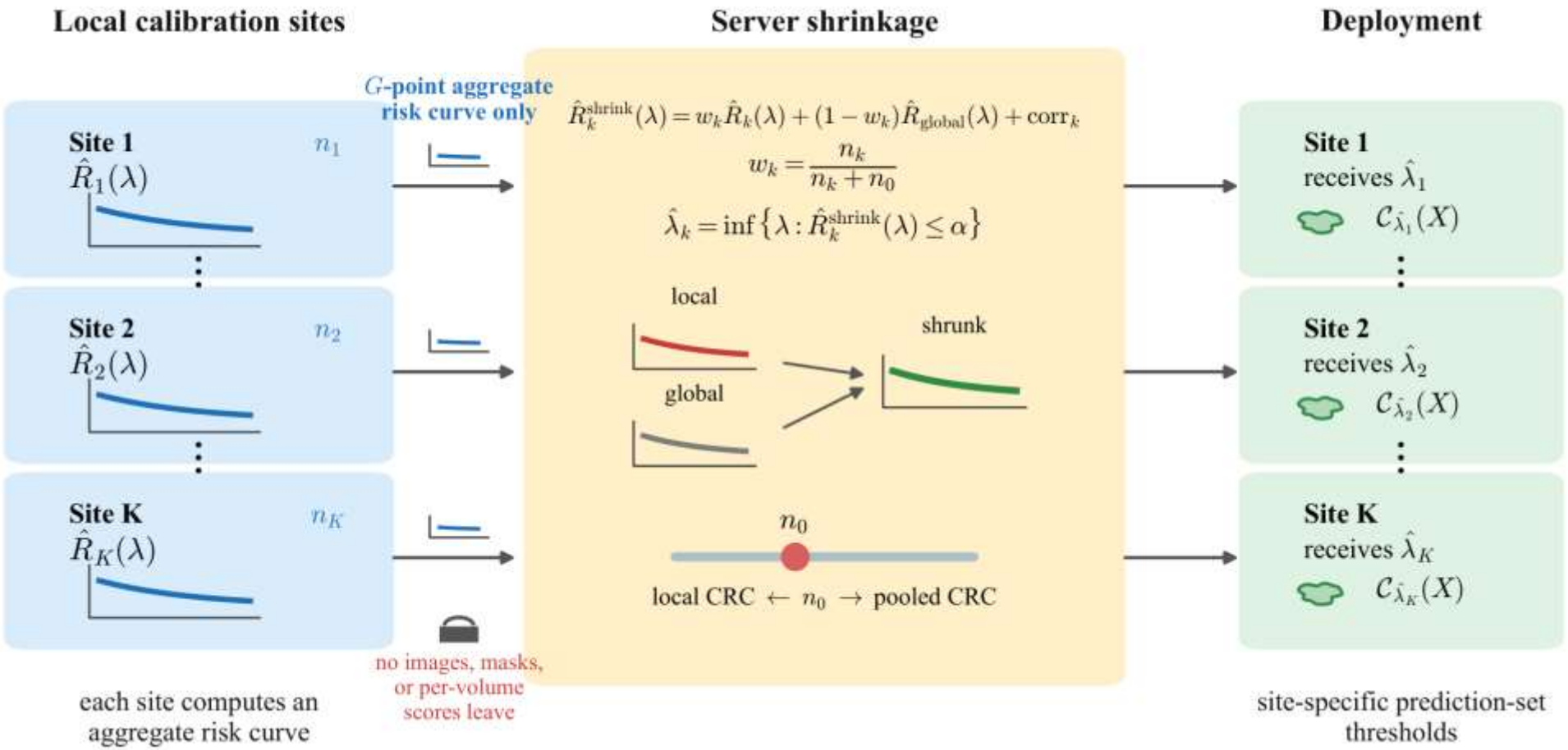


Fig. 1 (paper): protocol overview